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**Note that, in an effort to be consistent, we have updated the tables and charts in article 1 to reflect data through the end of May so that results are comparable with the results in this article.*

Introduction:

In article 1 (https://gestaltu.com/2013/05/dynamic-asset-allocation-for-practitioners-part-1-the-many-faces-of-momentum.html) of this series, we examined several ways of measuring the momentum factor in the context of a dynamic asset allocation framework. We presented results backed by years of data and found that some measures were superior, while others exhibited a slightly different character. This is largely attributed to the responsiveness of how different metrics react to new asset trends. In this second article in our series, we dig deeper into additional momentum metrics by measuring the momentum of asset classes against a variety of risk measures.
From an opportunity cost perspective, investors are faced with an assortment of return streams and the asset allocator must intelligently select combinations to maximize return and minimize risk. This article will allow us to compare methods of selecting assets based on risk adjusted momentum to methods that select assets based on raw momentum measures.
Methodology:
Similar to our first article series, we are going to be applying the same analytical framework with 13 different indicators. We aim to avoid presenting curve fitted results that look promising in sample but are unlikely to perform as expected out of sample.
More specifically, we take into account portfolio concentration and universe specification. The former refers to the performance of strategies holding the top N different assets at each rebalance. Rather than arbitrarily choose a top N based on what has worked best in the past, our results will average the results of holding between 2 and 5 top assets.
Universe specification bias is the potential bias arising from our preselected universe of assets. If the performance of a system is largely attributed to holding one or two lucky assets, future performance may be highly dependent on including the best performing asset in the future, which we may not know about today. To minimize this probability, we created a framework whereby performance is measured through sequentially dropping one asset at a time, and averaging the results across universes. This helps to minimize universe pre-selection bias.
Our 10 asset universe:

- Commodities (DB Liquid Commoties Index)
- Gold
- U.S. Stocks (Fama French top 30% by market capitalization)
- European Stocks (Stoxx 350 Index)
- Japanese Stocks (MSCI Japan)
- Emerging Market Stocks (MSCI EM)
- U.S. REITs (Dow Jones U.S. Real Estate Index)
- International REITs (Dow Jones Int'l Real Estate Index)
- Intermediate Treasuries (Barclays 7-10 Year Treasury Index)
- Long Treasuries (Barclays 20+ Year Treasury Index)

The following description from our first article expands on the technical methodology:
"For each strategy we will show the average statistics for all simulations with 2, 3, 4, and 5 holdings, and across all 11 asset universes. Recall that we are testing the full 10 asset class universe, as well as 10 other 9 asset class universes where one of the original assets is removed. So the statistics for each strategy will actually represent an average (median) of 44 simulations (4 portfolio concentrations x 11 universes). We will then present modified histograms to illustrate the range of outcomes for each strategy."
Momentum Metrics
We are going to present 13 different risk adjusted momentum measures and test them over the period 1995 – present. Each different indicator, while correlated, differs in their way of measuring risk and return. While on average the assets held are similar, there are instances where certain metrics will yield different holdings, providing an opportunity for diversification across methodologies.
The following are the short descriptions and formulas:

- Sharpe Ratio- Created by William Sharpe, it is a popular risk adjusted ratio that measures excess return per unit of volatility, according to the following formula:

$SharpeRatio = \frac{R - R_f}{\sigma_R}$ (http://4.bp.blogspot.com/-uMMhLGuy1HY/UcHH3hUEWXI/AAAAAAAABx8/pMqZc7_6AnM/s1600/Sharpe+Ratio.gif)

- Omega Ratio- The Omega Ratio is a measure that takes into account all the different moments of the distribution. It separates return above and below a given threshold before calculating a ratio between the two means of the two vectors.

$\Omega = \frac{\int_r^\infty 1 - F(x)dx}{\int_\infty^r F(x)dx}$ (http://1.bp.blogspot.com/-cH3vGpP4UcA/UcHH26iU18I/AAAAAAAABxo/pxGwuCb0ykE/s1600/Omega.gif)

- Sortino Ratio- The Sortino Ratio is a modified version of the Sharpe ratio. While the Sharpe ratio takes into account both upside and downside volatility the Sortino ratio only uses the downside semi deviation in the denominator. The intuition behind this measure is that investors don't penalize upside volatility, so the risk measure should only focus on downside risk.

$Sortino = \frac{R - R_f}{DR_R}$ (http://3.bp.blogspot.com/-vf_Sebq8kxg/UcHH3t1c7XI/AAAAAAAAByc/FQbx9QTce24/s1600/Sortino.gif)

DR simply represents the downside semi-deviation.

- Calmar Ratio (MAR)- Published by the Managed Accounts Reports, this risk return measure uses the largest drawdown as its proxy for risk. It is widely used to gauge an investment's annualized return to its maximum drawdown.

$MAR = \frac{CAGR}{MaxDrawdown}$ (http://1.bp.blogspot.com/--WUePH6m_tM/UcHH2dkVceI/AAAAAAAABxQ/VayQYdkaDIk/s1600/MAR.gif)

- DVR- This ratio, which was formalized by David Varadi, is concerned with both returns per unit of risk, and the linear fit of the price trajectory. Technically, it is the product of the Sharpe ratio and the coefficient of determination, or R-squared measure, fit to the price trajectory through time.

$DVR = SharpeRatio \times R^2$ (http://2.bp.blogspot.com/-f4UIVfTeYYQ/UcHH1y9y3II/AAAAAAAABw8/pEwBrr-9sdQ/s1600/DVR.gif)

- Value at Risk- Used widely in the financial industry for measuring firm wide risk and exposure, the Value at Risk calculates, for a given confidence level, the magnitude of expected loss. We took this risk measure and incorporated the mean return to derive a risk return ratio.

$\vartheta_{VaR} = \frac{\bar{R}}{VaR_\alpha}$ (http://2.bp.blogspot.com/-k2oUdgzxy8o/UcHH3UehSzI/AAAAAAAABx4/2Yb21VBnffs/s1600/SF-VaR.gif)

- Conditional Value at Risk- A variation of the VaR, the Conditional Value at Risk is simply the median value of all return observations between the maximum loss and the VaR for a given confidence level. Theoretically, it does a better job of capturing the true tail risk. Similar to the above, we adjusted it for the mean return for a risk return ratio.

$\vartheta_{CVaR} = \frac{\bar{R}}{CVaR_\alpha}$ (http://1.bp.blogspot.com/-I9A5TIfpDfM/UcHH3EKFqMI/AAAAAAAAByA/wvjaL2II8L8/s1600/SF-CVaR.gif)

- Return to Max Loss Ratio- The max loss ratio uses the worst daily return as a proxy for risk.

$MaxLossRatio = \frac{\bar{R}}{min(R_D)}$ (http://4.bp.blogspot.com/-BcpfmMELK7g/UcHH2tItxuI/AAAAAAAABxg/dZDdoa46mzc/s1600/MaxLoss+Ratio.gif)

$min(R_D) = \begin{bmatrix} R_1 & R_2 & ... & R_t \end{bmatrix}$ (http://4.bp.blogspot.com/-jyKSyZVpq00/UcHH2TJJtCI/AAAAAAAAByE/019GdkPOx6s/s1600/MaxLoss+Ratio+2.gif)

- Return to Average Drawdown Ratio- The Average drawdown ratio simply takes return and adjust it for the average drawdown over the entire period.

$AverageDrawdownRatio = \frac{\bar{R}}{AverageDrawdown}$ (http://2.bp.blogspot.com/-ow7p3UHBfD8/UcHH19QgA0I/AAAAAAAABxA/6F4G63vATik/s1600/AverageDrawdown.gif)

- High Low Differential- More sophisticated in calculation, the High Low Differential takes in to account the current position relative to the highest and lowest prices in a prescribed period.

$\Delta_{high} = \frac{max(price_{1...t})}{price_t}$ (http://1.bp.blogspot.com/-y44iC-qF-NQ/UcHH37AXDPI/AAAAAAAAByU/S0ARZbqB10o/s1600/DeltaHigh.gif)
$\Delta_{low} = \frac{price_t}{min(price_{1...t})}$ (http://1.bp.blogspot.com/-gYks3cpw22c/UcHH4YX8KMI/AAAAAAAAByw/WsuJHhEJDFs/s1600/DeltaLow.gif)
$Differential_{High} = \frac{\Delta_{high}}{\sigma}$ (http://3.bp.blogspot.com/-7AkaI6OiyZc/UcHH4bwifUI/AAAAAAAABy4/pHkLzgt3Amw/s1600/HighDifferential.gif)
$Differential_{Low} = \frac{\Delta_{low}}{\sigma}$ (http://4.bp.blogspot.com/-jTyeEIehu-s/UcHH40KA0EI/AAAAAAAABy8/oUooxRrgZvU/s1600/LowDifferential.gif)
$Differential_{LowHigh} = \frac{Differential_{Low} - Differential_{High}}{100}$
(http://4.bp.blogspot.com/-1uZOKS4vl1s/UcHH4pthUmI/AAAAAAAABys/kusi69oj7CI/s1600/LHDiff.gif)

- Ulcer Index- Created by Peter Martin in 1987, the index takes into account risk from drawdowns as oppose to traditional volatility. It is derived from deviations from the most recent highs. The following is the pseudocode for computing the Ulcer Index (UI), the following is the equation employing the UI to derive the UPI.

```
  SumSq = 0
MaxValue = 0
for T = 1 to NumOfPeriods do
    if Value[T] > MaxValue then MaxValue = Value[T]
else SumSq = SumSq + sqr(100 * ((Value[T] / MaxValue) – 1))
UI = sqrt(SumSq / NumOfPeriods)
```

Source: http://www.tangotools.com/ui/ui.htm

$UPI = \frac{\bar{R} - R_f}{UI}$ (http://3.bp.blogspot.com/-4j3IYBXOE4E/UcHH3853XZI/AAAAAAAAByY/UpfkzblJKM0/s1600/UPI.gif)

- Gain to Pain Ratio- Popularized in the book Hedge Fund Market Wizards by Jack Schwager, this ratio takes the sum of all positive periods divided by the sum of all the negative periods.

$GPR = \frac{\sum x_{positive}}{|\sum x_{negative}|}$ (http://4.bp.blogspot.com/-Qui9Iz3Mtys/UcHH2KD2N_I/AAAAAAAABxY/7lpU1zH8D60/s1600/GPR.gif)

- Fractal Efficiency- The most efficient line segment between two points is a straight line. Essentially, fractal efficiency is the ratio between the straight-line magnitude of price change over the period divided by the distance the price actually traveled on its path. The equation below should help with the intuition.

$FractalEfficiency = \sum |L_i - Price_i|$ (http://1.bp.blogspot.com/-mS4uj3M3Qro/UcHH18YAG6I/AAAAAAAABxI/UQBMUEkuHK8/s1600/FractalEfficencyV2.gif)

Like our last post, we have applied the same transformation to standardize the momentum measures to avoid temporal issues from employing multiple lookback horizons. Again, our standardization equation is:

$StandardizedMomentum = \frac{|momentum_i|}{\sum |momentum_{i..n}|} \times sign(momentum_i)$ (http://1.bp.blogspot.com/-lA2uK6Fvwuc/UcHJ3PYOyXI/AAAAAAAABzI/BLrslVa7UOI/s1600/standardization2.jpg)

Results

The median performance table (Table 1.) summarizes the summary statistics of our 13 momentum risk adjusted systems. Results represent the median across all combinations of varying concentrations and universes, so each statistic summarizes performance across 44 individual tests. Among the top contenders we have the DVR and return / Max Loss Ratio from a risk-adjusted basis (Sharpe), and the return / Ulcer Index (UPI) indicator delivers the best return / maximum drawdown (MAR).

Table 1. Median Performance Summary

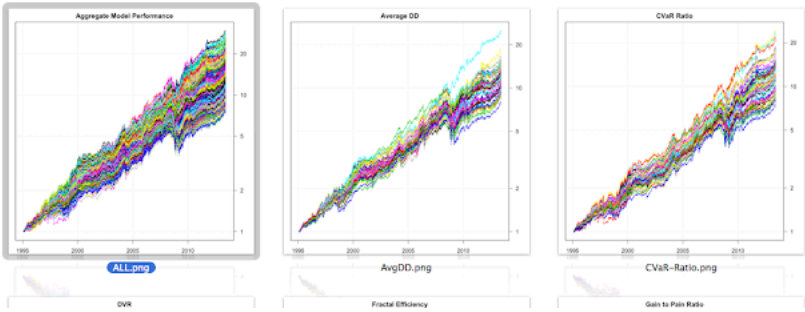
	Sharpe	Omega	Sortino	MAR	DVR	VaR-Ratio	CVaR-Ratio	MaxLoss Ratio	Avg DD	LoHi Diff	Fract. Eff.	UPI Ratio	GP Ratio
CAGR	15.03	14.975	15.005	14.65	15.875	15.33	14.18	14.985	14.07	15.415	14.9	15.545	14.75
Sharpe (2%)	1.175	1.175	1.19	1.115	1.235	1.21	1.17	1.225	1.165	1.195	1.165	1.21	1.205
MaxDD	-29.79	-24.005	-23.29	-32.08	-23.54	-23.835	-23.34	-23.34	-27.575	-30.6	-26.585	-23.44	-24.005
MAR	0.4739	0.59015	0.65895	0.46105	0.6605	0.6382	0.60565	0.6339	0.5387	0.49445	0.5482	0.6811	0.59235
Volatility	12.315	12.24	12.045	12.265	11.995	12.15	12.125	12.175	12.05	12.1	12.16	12.03	11.9
Positive 12 Month	93.515	92.64	94.415	93.225	93.665	95.335	94.59	94.14	93.545	93.675	92.925	95.185	93.065

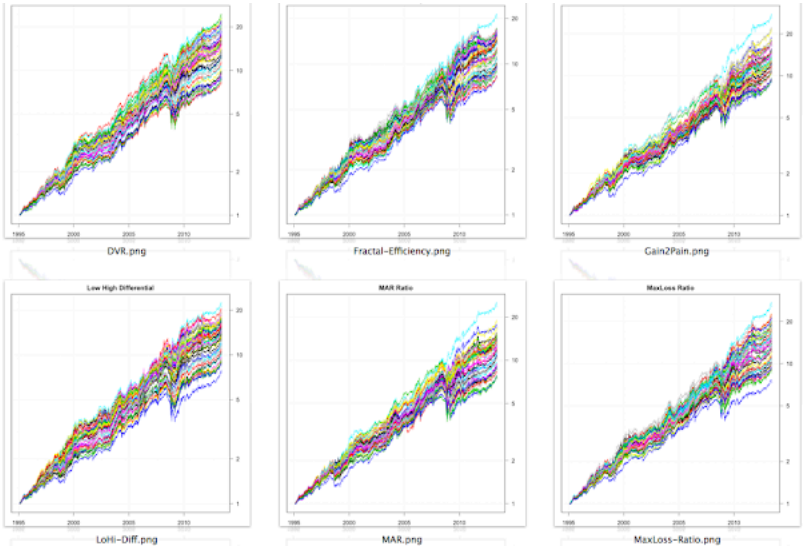
(http://1.bp.blogspot.com/-YelMM57S_tQ/UcHOfoJNTDI/AAAAAAAABzY/Kqk0FPjUmQ8/s1600/Median+Performance+Chart1.png)

Source: Bloomberg

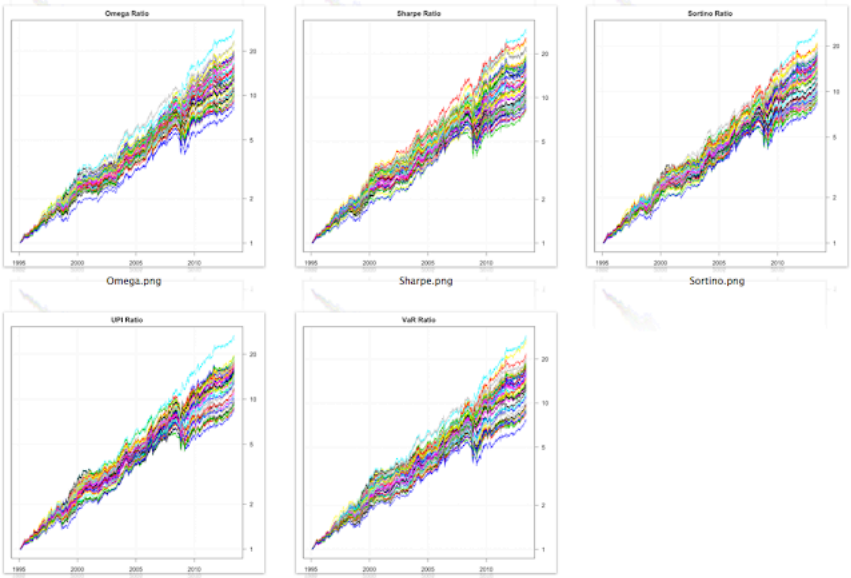
Perhaps counter-intuitively, the risk adjusted momentum portfolios exhibit lower Sharpe ratios than the raw momentum systems tested in Article 1. As you will see, it may be more coherent, and produce better results, to disaggregate the application of risk management from the momentum measure.

Chart 1-14 Performance Distributions.





(http://4.bp.blogspot.com/-uPL1Xt4cp-w/UclhOXpPbLI/AAAAAAAAB2c/Jw8vkMPBalY/s1600/Equity_Lines_1.png)



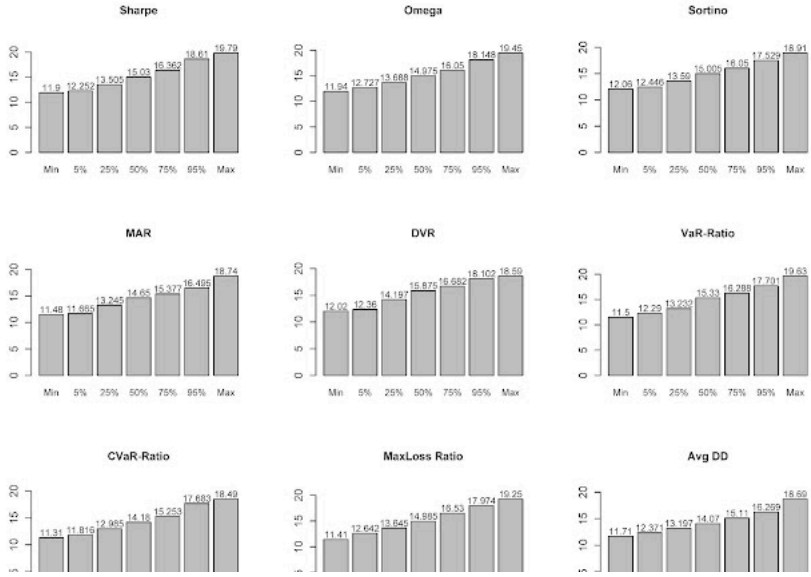
(http://4.bp.blogspot.com/-9ml8IGv3KQs/UclhNuEcv1I/AAAAAAAAB2U/691OMme7pHI/s1600/Equity_Lines_2.png)

Source: Bloomberg

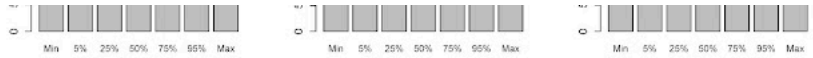
Charts 1-14 display the equity line distributions from varying concentration and universes. Visualizing the equity lines in such a way allows one to identify the consistency from all the different combinations. The robustness of a system is measured as a function of how it responds to changing environments and parameters. Methods that are resilient to different universes and portfolio concentrations are likely to deliver more stable results out of sample.

A cursory glance reveals that the Low-High Differential shows the most variability while the Gain to Pain Ratio (debatable) shows the least.

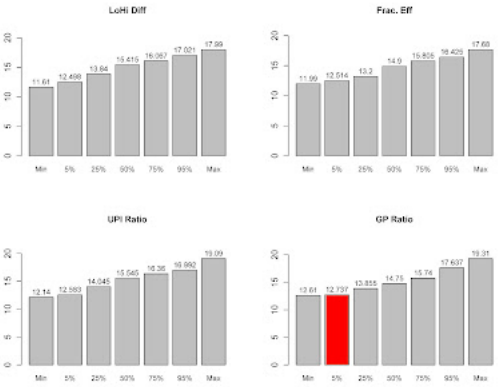
Chart 15-54 Charts (CAGR)



(http://3.bp.blogspot.com/-

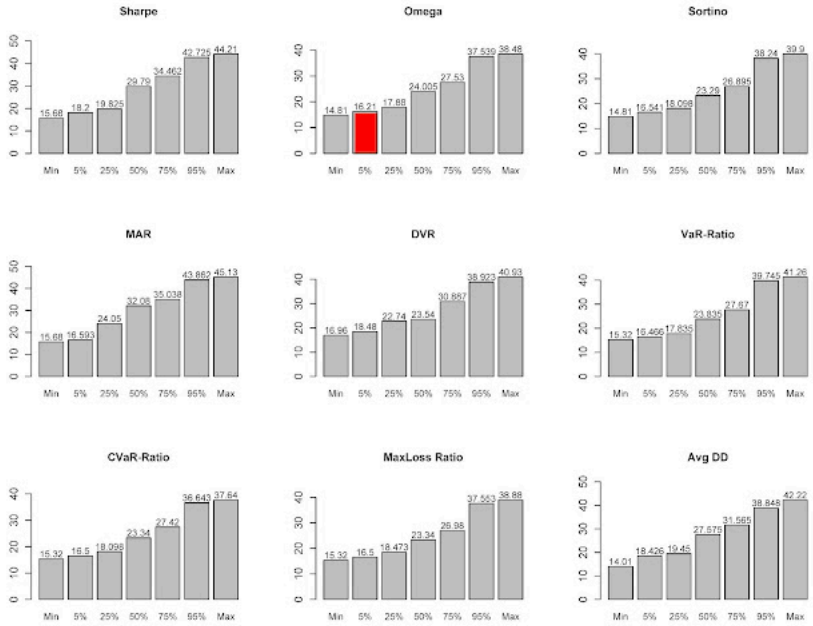


haOyZVm5hUY/UcHuKrb76YI/AAAAAAAAB1k/ae-rBBsOqvU/s1600/Quantile-CAGR11.png)



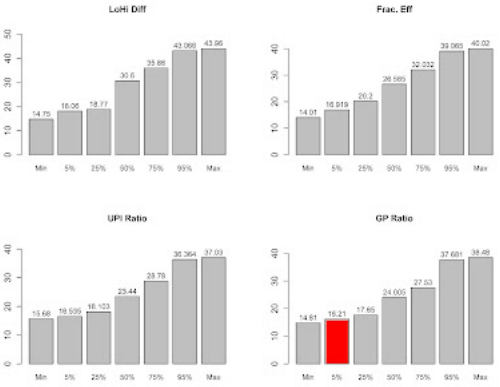
(http://4.bp.blogspot.com/-

BFeIdrNaU2w/UcHuKhlz3bI/AAAAAAAAB1o/egDzgR9T3nM/s1600/Quantile-CAGR21.png)
Source: Bloomberg
Max Drawdown



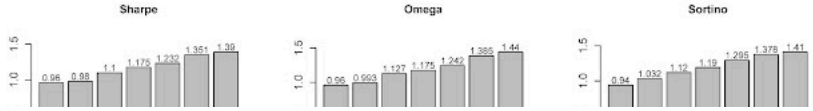
(http://3.bp.blogspot.com/-kDu6ErNUQxg/UcHuN-

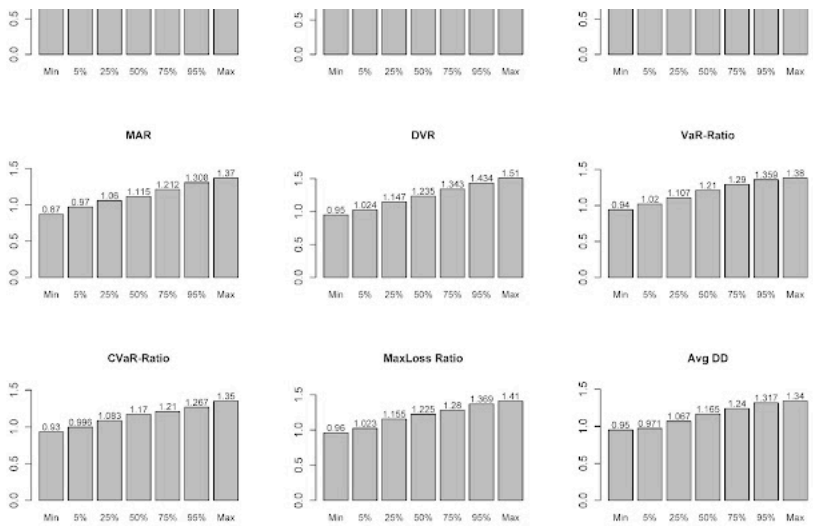
LGA_I/AAAAAAAAB1w/fJfEGiF4Mow/s1600/Quantile-MaxDD11.png)



(http://1.bp.blogspot.com/-

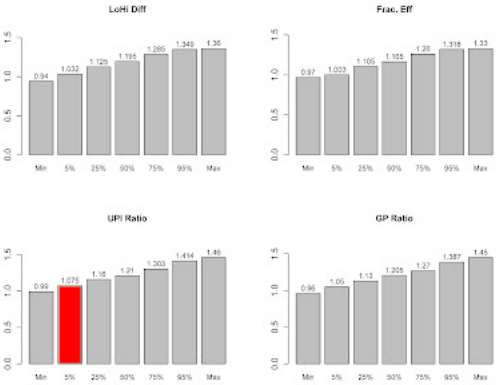
sz6lsSy_izo/UcHuOu04LLI/AAAAAAAAB14/oJPs1RAr4kk/s1600/Quantile-MaxDD21.png)
Source: Bloomberg
Sharpe





(http://1.bp.blogspot.com/--

Aat0SsR8Bs/UcHuPw6rPWI/AAAAAAAAB2A/CKMhTmeLsXo/s1600/Quantile-Sharpe11.png)



(http://1.bp.blogspot.com/-

r6i8PLqHzaw/UcHuQ7G3muI/AAAAAAAAB2I/dtHRcdktans/s1600/Quantile-Sharpe21.png)

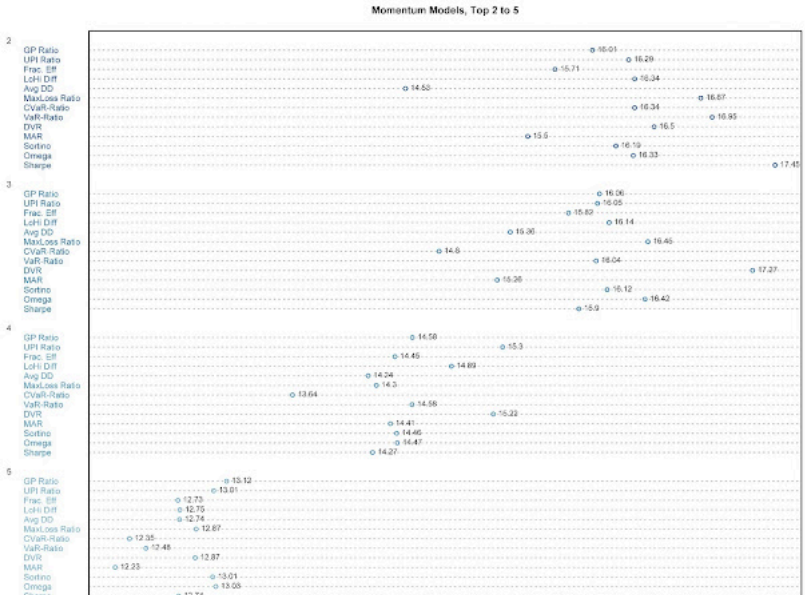
Source: Bloomberg

Consistent with our previous post's analytical framework, we show the distribution of performance statistics across all of the 44 universe/concentration combinations for each risk adjusted momentum method.

Below we plot the percentile performance of each system's CAGR, Sharpe, and Maximum Drawdown, paying special attention to scores at the 5th percentile, because this quantile is standard for interpreting statistical significance. Whereas the instantaneous slope measure proved to be the most consistent performer at the 5th percentile among raw momentum metrics in Article 1, the Gain to Pain ratio seems to deliver the most consistent performance of all the risk adjusted momentum indicators.

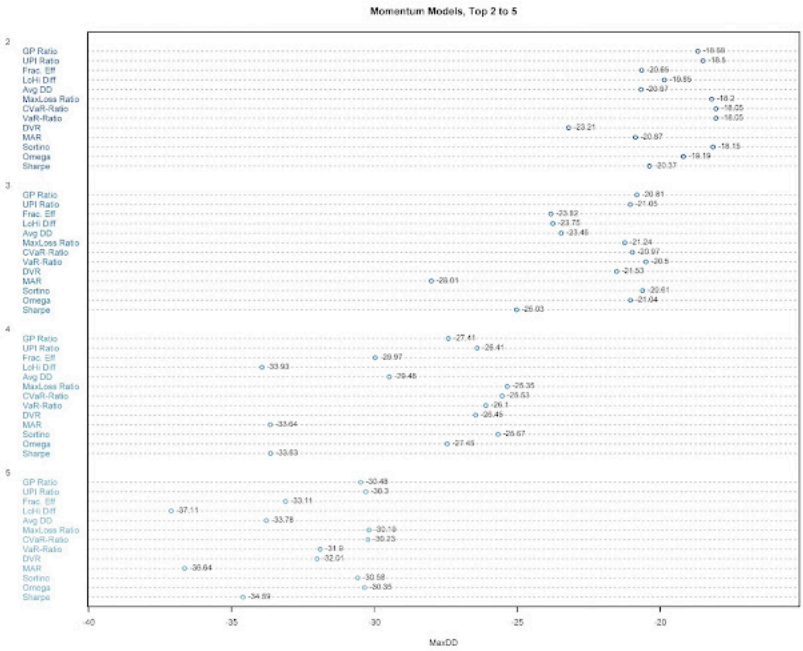
Charts 55 through 57 show the average performance of each methodology with portfolio concentrations of 2 holdings through 5 holdings across all 11 asset universes tested.

Chart 55: CAGR



(http://2.bp.blogspot.com/-

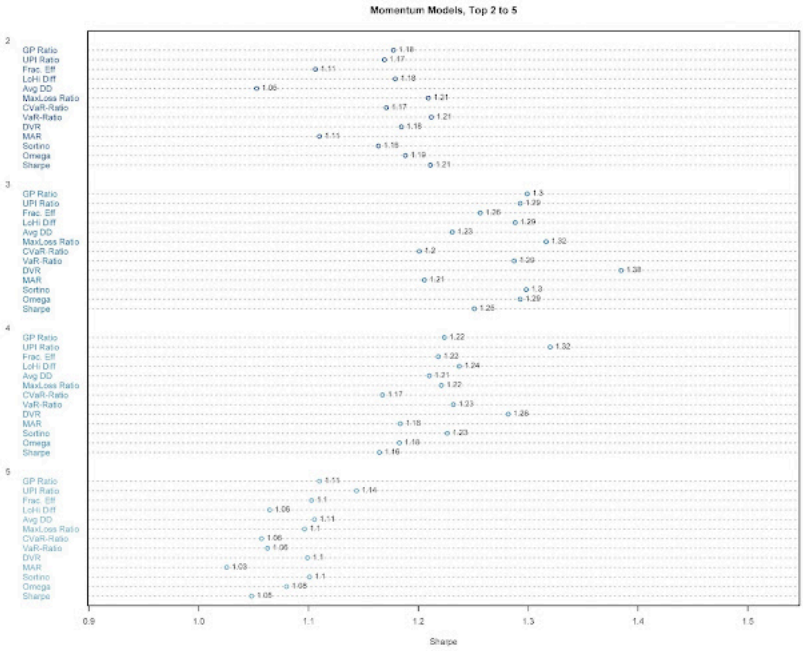
cCutN6ymxng/UcHewzcufjI/AAAAAAAAB0Y/E03O6duVaqI/s1600/CAGR-DotChart1.png)
Source: Bloomberg
Chart 56: Max Drawdown



(http://2.bp.blogspot.com/-

KEJdV72PoYw/UcHex_fyj5I/AAAAAAAAB0o/mpEQBSPxfkk/s1600/MaxDD-DotChart1.png)

Source: Bloomberg
Chart 57. Return / risk ratio



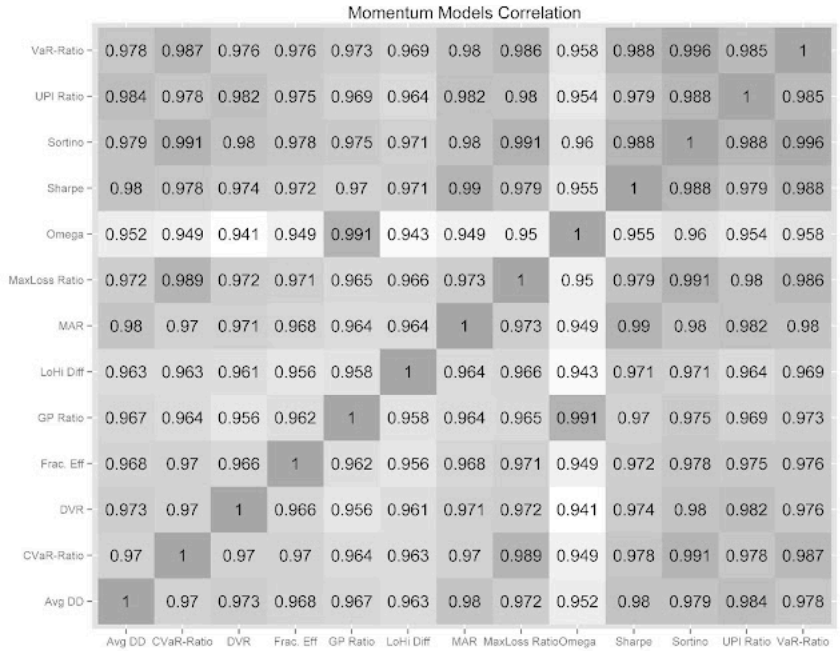
(http://3.bp.blogspot.com/-

ri3mgTTzd4w/UcHexUI_XfI/AAAAAAAAB0c/HckaHvfj1wQ/s1600/Sharpe-DotChart1.png)

Source: Bloomberg
Consistent with observations from Article 1., while more concentrated portfolios tend to deliver higher returns, the highest Sharpe ratios are derived from portfolios with 3 or 4 holdings. However, in contrast with the raw momentum tests, risk adjusted portfolios seem to deliver monotonically smaller drawdowns moving from 2 holdings to 5 holdings.

Indicator Diversification

Any single strategy by itself suffers its own structural inadequacies because no individual indicator or system effectively captures all of the information that is available in the price series.
Chart 58 Correlation Matrix

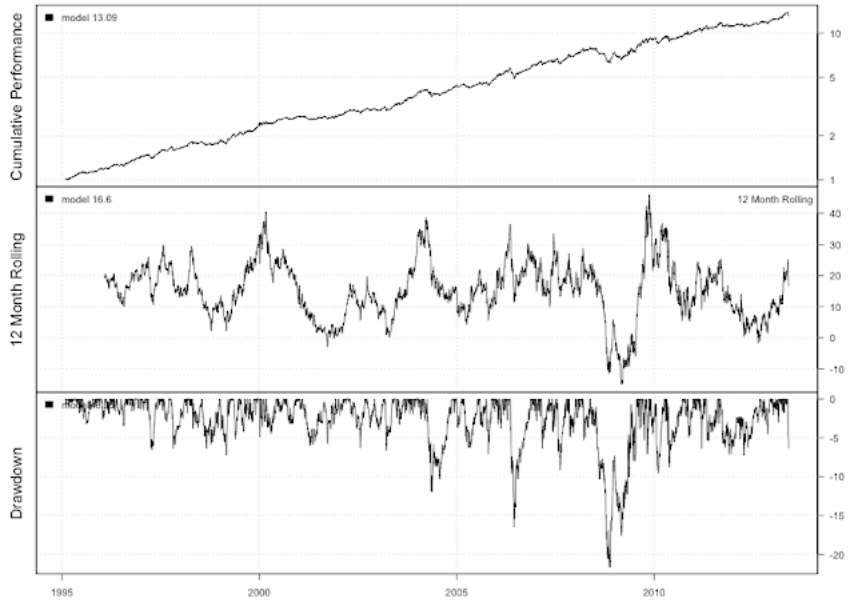


(http://1.bp.blogspot.com/-nybviYY3IR0/UcHfWsvU4vI/AAAAAAAAB00/QD6bgIbnxFc/s1600/Correlation-Matrix1.png)

Source: Bloomberg

Although the correlations between the systems are highly correlated, they only capture the long-term average correlation over the entire period. If one were to view them on a rolling basis, they will identify periods where their returns diverge. To prove this point we aggregated the systems all together in an equal weight index.

Chart 59 and Table 2.



(http://4.bp.blogspot.com/-2elupiUn5Qg/UcHfgT0jjDI/AAAAAAAAB08/R4VjYrYpOkk/s1600/Average-Index+Equity.png)

	System		Trade		Period
Period	Feb1995 - May2013			Win.Percent.Day	55.9
Cagr	15.06			Win.Percent.Month	65.5
Sharpe	1.33			Best.Month	10.8
DVR	1.22			Worst.Month	-10.2
Volatility	10.98			Win.Percent.Year	94.7
MaxDD	-21.65			Best.Year	33.1
AvgDD	1.7			Worst.Year	0.6

(http://3.bp.blogspot.com/-fCi0Gbds2Dw/UcHjFe-

AvgDD	-1.7			Worst Year	0.0
VaR	-1.07				
CVaR	-1.6				
Pos.12Month	95.71				

KIWI/AAAAAAAAB1M/PyxtkoHLJdY/s1600/Rplot1.png)
Source: Bloomberg

Comparing the index to all the constituent systems from Table 1., we observe a material reduction in volatility and drawdown. Among the improved performance statistics include Sharpe, Maximum Drawdown, and rolling positive 12 month periods. Clearly the different measures of risk capture slightly different information, which offers diversification at more critical times.

Conclusion

Articles 1. and 2. have analyzed the subject of momentum superficially by introducing myriad indicators to measure the momentum anomaly. While they differ in their calculation, all of them measure the strength of recent 1 to 12 month absolute or risk-adjusted price strength. To now, we have held all assets in equal weight to focus attention purely on the momentum metric.

In upcoming posts, we will introduce a variety of sizing algorithms. We will incorporate both traditional optimization procedures and heuristic methods to identify the optimal sizing combination. Stay tuned.

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